Sensor networks

Joint Clustering and 3-D UAV Deployment for Delay-Aware UAV-Enabled MTC Data Collection Networks

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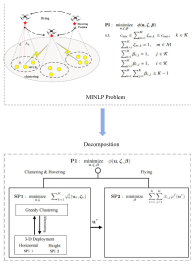
Abstract—The design of timely data collection for a machine-type communication (MTC) network by unmanned-aerial-vehicle (UAV) platform is investigated. The ground-based MTC devices are clustered for efficient service, and the UAV station's deployment in the 3-D space is optimized. The corresponding mission time minimization problem is formulated as a coupled mixed-integer nonlinear program. For tractability, the original problem is decomposed into two subproblems respectively dealing with clustering-hovering optimization and intercluster UAV traveling path minimization. An alternating clustering-hovering optimization (ACH) and ant colony optimization (ACO) solution approach is proposed accordingly. Simulations are conducted to validate the superiority of the proposed ACH–ACO scheme over the scheme based on k-means clustering.



Index Terms—Sensor networks, clustering, mission time minimization, UAV 3-D deployment, UAV-assisted data collection.

I. INTRODUCTION

As a key foundation of Internet-of-Things (IoT) applications, machine-type communication (MTC) has been developing rapidly in recent years [1]. By deploying a large number of machine-type communication devices (MTCDs) to sense the environment and exchange control messages, MTC networks can effectively support timely intelligent management applications of information and physical systems, such as autonomous driving and factory automation [2]. Central to quality-of-service provisioning in such applications of MTC networks is efficient data collection from a large number of MTCDs with latency considerations [3].

In conventional data collection schemes, the MTCDs transmit data to a ground-based collector station directly or through relaying, which is easily blocked by obstacles and energy-inefficient. To overcome the above issues, UAVs have been introduced to establish airborne data sinks to support more efficient and flexible data collection [4], [5]. Owing to its high mobility, a UAV can approach ground MTCDs and provide strong Line-of-Sight (LoS) links. As a result, UAV-assisted data collection can achieve much lower mission time than conventional schemes based on fixed ground stations.

There have been many works in the literature aiming at reducing latency of UAV-assisted data collection through trajectory optimization. For instance, in [6], trajectory optimization and data collection algorithms were proposed for simultaneous shortening flight time and improving data collection rate. By jointly optimizing the UAV station's trajectory, flying height, and velocity, a UAV-aided data collection design was proposed in [7] to minimize the total mission time. However, it is challenging to accurately model the influence of the highly dynamic UAV flight attitude on wireless transmission, so the above works ignore the impact of UAV flight attitude on communication. Noting that the attitude of the UAV is easy to control while hovering, [8] and [9] proposed to collect data at hovering locations on the trajectory, and the optimal trajectory was designed with such practical constraints taken into account.

On the other hand, due to the limited coverage and capacity of UAV stations, it is reasonable to serve a large number of MTCDs in a clustered manner. However, joint design of MTCD clustering and UAV hovering is a complex coupled optimization problem, which has yet been adequately studied. In [10], the ground MTCDs were first clustered, the UAV then hovers above each cluster during its flight for data collection. Since the UAV hovering positions are obtained through weighted k-means principle, the scheme has no optimality guarantee. A dual-hop design that used a cluster head to gather data from close-by MTCDs and then forwarded it to the UAV station was investigated in [11]. This two-hop process had a significant additional overhead, and hence is undesirable for IoT applications.

In this letter, we consider minimization of the overall mission time in a UAV-assisted MTC data collection network by jointly optimizing MTCD clustering, UAV hovering, and UAV flying strategies in a 3-D space. The problem is formulated as a coupled mixed-integer nonlinear program (MINLP). To tackle the coupled issue, the problem is separated into two subproblems, which respectively optimize the clustering-hovering strategy and the intercluster flying strategy. The former determines the communication/hovering time, and the latter determines the traveling time. An alternating clustering-hovering optimization and ant colony optimization (ACH–ACO) solution approach is proposed. Specifically, ACH is responsible for clustering-hovering optimization, and ACO designs intercluster flying trajectories. Through numerical examples, the proposed scheme is shown to effectively reduce the overall mission time.

II. SYSTEM MODEL

As shown in Fig. 1, a UAV-enabled data collection system is deployed for a MTC network. One UAV collects data from M

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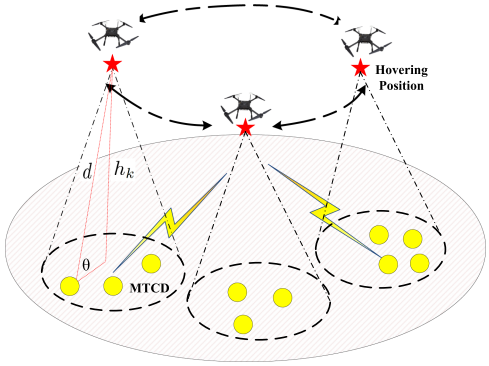

Fig. 1. UAV-enabled data collection system model.

static ground-based MTCDs while flying over the target service area.

The MTCD set is denoted as $\mathcal{M} \triangleq \{1, 2, \ldots, M\}$. Assume that the UAV has acquired all the MTCDs' position information in advance, which is a fair assumption considering only sub-meter accuracy is required. The UAV has limited coverage and service capacity. Therefore, the MTCDs should be divided into multiple clusters based on MTCD distribution density and the UAV's capabilities [9]. Assuming the cluster number is K, the UAV travels across the clusters and collects data from each cluster while hovering above it. The cluster index set is denoted as $\mathcal{K} \triangleq \{1, 2, \ldots, K\}$. The altitude of the ground is set to 0. The horizontal position coordinates of the MTCDs in cluster k are expressed as $\mathbf{s}_m = (s_m^x, s_m^y)$, $m \in \mathcal{M}$. Similarity, the 3-D coordinates of the hovering position of the UAV for collecting data from the kth cluster is expressed as $\mathbf{u}_k = (u_k^x, u_k^y, h_k)$. To ensure stable reception of the UAV station, we make a practical assumption that no data communication is allowed while the UAV is flying between hovering positions.

The air-to-ground link can be modeled by considering the occurrence probability of either LoS condition or nonline-of-sight (NLoS) condition. According to [12], the pathlosses of the LoS and NLoS links are expressed as

$$PL_{\mathrm{L}} = 20\lg d + 20\lg f + 20\lg(4\pi/c) + \Delta_{\mathrm{LoS}}, \tag{1}$$

$$PL_{\mathrm{NL}} = 20\lg d + 20\lg f + 20\lg(4\pi/c) + \Delta_{\mathrm{NLoS}} \tag{2}$$

where d is the distance between the transmitter and the receiver, and Δ_{LoS} and Δ_{NLoS} are the corresponding additional pathlosses in dB on top of the free-space propagation pathloss, respectively.

The LoS occurrence probability is

$$\Pr_{\mathrm{L}}(\theta) = 1/\left(1 + a\exp\left(-b\left[\theta - a\right]\right)\right) \tag{3}$$

where a and b are constants related to the propagation environment, and $\theta = \arcsin(\frac{h_k}{d})$ is the elevation angle.

The average achievable rate between the UAV hovering above the kth cluster and MTCD $m \in \mathcal{M}$ is given by

$$\begin{aligned} R_m(\mathbf{u}_k) = {} & \Pr_{\mathrm{L}}(\theta_m(\mathbf{u}_k))\log_2\left(1 + p_m/(\lambda_{\mathrm{L}}(\mathbf{u}_k)\sigma^2 d(\mathbf{u}_k)^2)\right) \\ & + (1 - \Pr_{\mathrm{L}}(\theta_m(\mathbf{u}_k))\log_2\left(1 + p_m/(\lambda_{\mathrm{NL}}(\mathbf{u}_k)\sigma^2 d(\mathbf{u}_k)^2)\right) \end{aligned} \tag{4}$$

where p_m is the transmit power of MTCD m, $\lambda_{\mathrm{L}}(\mathbf{u}_k) = 10^{\frac{PL_{\mathrm{L}}}{10}}$, $\lambda_{\mathrm{NL}}(\mathbf{u}_k) = 10^{\frac{PL_{\mathrm{NL}}}{10}}$, and $d(\mathbf{u}_k)^2 = \|\mathbf{u}_k - \mathbf{s}_m\|_2^2$.

III. PROBLEM FORMULATION

Under the premise of completing all data collection tasks, the overall mission time is minimized by jointly optimizing the MTCD clustering and UAV's trajectory. Denote by η_m the data size that MTCD m needs to upload. The hovering time for collecting data from MTCD m is measured by

$$t_m(\mathbf{u}_k) = \eta_m / R_m(\mathbf{u}_k). \tag{5}$$

Let $\boldsymbol{\zeta}_k \triangleq (\zeta_{m,k})_{m\in\mathcal{M}}$ represent the clustering strategy, where $\zeta_{m,k} \in \{1, 0\}$ is an indicator variable denoting whether the MTCD m belongs to the kth cluster. To ensure that all MTCDs finish uploading their data, the UAV hovering time above the kth cluster should be at least

$$\varphi_k^h(\mathbf{u}_k, \boldsymbol{\zeta}_k) = \max_{m\in\mathcal{M}}\{\zeta_{m,k} t_m(\mathbf{u}_k)\}, k \in \mathcal{K}. \tag{6}$$

Indicator variables $\beta_{i,j} \in \{1, 0\}$ are to denote whether the path from hovering position i to j is selected. The flying time of the UAV from the ith hovering position to the jth hovering position is

$$\varphi^f(\mathbf{u}_i, \mathbf{u}_j) = \left\|\mathbf{u}_i - \mathbf{u}_j\right\|_2 / v, i \in \mathcal{K}, j \in \mathcal{K} \tag{7}$$

where v is the maximum flying speed of the UAV.

For convenience, we define $\boldsymbol{\beta} \triangleq (\beta_{i,j})_{i,j\in\mathcal{K}}$, $\boldsymbol{\zeta} \triangleq (\boldsymbol{\zeta}_k)_{k\in\mathcal{K}}$ and $\mathbf{u} \triangleq (\mathbf{u}_k)_{k\in\mathcal{K}}$. The total mission time therefore becomes

$$\phi(\mathbf{u}, \boldsymbol{\zeta}, \boldsymbol{\beta}) = \sum_{k=1}^{K}\varphi_k^h(\mathbf{u}_k, \boldsymbol{\zeta}_k) + \sum_{i=1}^{K}\sum_{j=1}^{K}\beta_{i,j}\varphi^f\left(\mathbf{u}_i, \mathbf{u}_j\right). \tag{8}$$

The problem that jointly optimizes the MTCD clustering and 3-D UAV deployment is formulated as

$$\mathbf{P1}: \quad \underset{\mathbf{u},\boldsymbol{\zeta},\boldsymbol{\beta}}{\text{minimize}} \quad \phi(\mathbf{u}, \boldsymbol{\zeta}, \boldsymbol{\beta}) \tag{9}$$

$$\text{s.t.} \quad c_{\min} \le \sum_{m=1}^{M}\zeta_{m,k} \le c_{\max}, \quad k \in \mathcal{K} \tag{9a}$$

$$\sum_{k=1}^{K}\zeta_{m,k} = 1, \quad m \in \mathcal{M} \tag{9b}$$

$$\sum_{i=1}^{K}\beta_{i,j} = 1, \quad j \in \mathcal{K} \tag{9c}$$

$$\sum_{j=1}^{K}\beta_{i,j} = 1, \quad i \in \mathcal{K} \tag{9d}$$

$$\sum_{i=1}^{K}\sum_{j=1}^{K}\beta_{i,j} \le K - 1 \tag{9e}$$

$$H_l \le h_k \le H_u, \quad k \in \mathcal{K}. \tag{9f}$$

Note that the inequality constraints (9a) and (9b) are introduced to set a reasonable range for the cluster size, which depends on the available orthogonal resource blocks. Constraints (9c), (9d), and (9e) are imposed to ensure that the UAV passes every hovering position only once and that no subloop paths exist. Constraint (9f) sets the upper and lower limits to the UAV flight altitude. Problem **P1** is a MINLP which is in general challenging to solve.

IV. PROPOSED CLUSTERING AND 3-D UAV DEPLOYMENT OPTIMIZATION STRATEGY

To tackle the MINLP problem **P1**, we decouple the clustering, hovering, and flying variables, such that the original problem is decomposed into two tractable subproblems.

A. Joint MTCD Clustering and UAV Hovering Optimization

First, we focus on the joint optimization of MTCD clustering and UAV hovering strategies to minimize the total hovering time. The

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corresponding subproblem is given by

$$\mathbf{SP1}: \underset{\mathbf{u},\boldsymbol{\zeta}}{\text{minimize}} \quad \sum_{k=1}^{K} \varphi_k^h(\mathbf{u}_k, \boldsymbol{\zeta}_k)$$
$$\text{s.t.} \quad (9a), (9b). \tag{10}$$

Problem **SP1** is still a challenging MINLP problem. We propose an iterative ACH algorithm that alternately optimizes the MTCD clustering variables $\boldsymbol{\zeta}$ that are integers and the UAV hovering variables $\mathbf{u}$ that are real numbers.

1) Optimizing the UAV Hovering Positions: For convenience, let $\mathcal{M}_k \triangleq \{m \in \mathcal{M} | \zeta_{m,k} = 1\}$ be the set of MTCDs in the kth cluster, for all $k \in \mathcal{K}$. Note that $\cup_{k\in\mathcal{K}}\mathcal{M}_k = \mathcal{M}$ and $\mathcal{M}_i \cap \mathcal{M}_j = \emptyset, \forall i \neq j, i \in \mathcal{K}, j \in \mathcal{K}$. Minimizing the hovering time of the UAV within a cluster is equivalent to maximizing the minimum data rate $R_{m_k}, m_k \in \mathcal{M}_k$. However, due to the complex nonconvex form of (4), it is challenging to obtain the optimal hovering position of the UAV.

Proposition 1: By introducing a lower bound on R_{m_k}, the 3-D UAV hovering position optimization is equivalently decomposed into horizontal position optimization and hovering height optimization.

Proof: It can be shown that $\log_2(1 + p_m/(\lambda(\mathbf{u}_k)\sigma^2 d(\mathbf{u}_k)^2))$ is a convex function of $\lambda(\mathbf{u}_k)$. The lower bound of the expected throughput can be derived based on Jensen's inequality

$$R_{m_k}(\mathbf{u}_k) \geq \log_2\left(1 + \frac{p_{m_k}\left(\frac{\Pr_{\mathrm{L}}(\theta_{m_k}(\mathbf{u}_k))}{\lambda_{\mathrm{L}}(\mathbf{u}_k)} + \frac{\Pr_{\mathrm{NL}}(\theta_{m_k}(\mathbf{u}_k))}{\lambda_{\mathrm{NL}}(\mathbf{u}_k)}\right)}{\sigma^2 d(\mathbf{u}_k)^2}\right)$$
$$\geq \log_2\left(1 + \frac{p_{m_k}}{\lambda(\theta_{m_k}(\mathbf{u}_k))\sigma^2 d(\mathbf{u}_k)^2}\right) \triangleq \bar{R}_{m_k}(\mathbf{u}_k). \tag{11}$$

First, the flying height is fixed and the horizontal coordinates of the UAV hovering position are optimized. The elevation angle of the MTCD m_k to the UAV is

$$\theta_{m_k}(\mathbf{u}_k) = \arcsin\left(h_k / \left\|\mathbf{u}_k - \mathbf{s}_{m_k}\right\|_2\right). \tag{12}$$

Combining (11) and (12), it is observed that the transmission rate between an MTCD and the UAV is a monotonically decreasing function of $\|\mathbf{u}_k - \mathbf{s}_{m_k}\|_2$. Therefore, the UAV hovering position optimization problem in the horizontal plane can be equivalently described as

$$\mathbf{SP1-1}: \underset{\{\mathbf{u}_k,\chi\}}{\text{minimize}} \quad \chi \tag{13}$$
$$\text{s.t.} \quad \left\|\mathbf{u}_k - \mathbf{s}_{m_k}\right\|_2^2 \leq \chi, \; m_k \in \mathcal{M}_k \tag{13a}$$

where χ is an auxiliary variable for construction of a standard convex problem. The problem (13) can be easily solved by CVX.

Then, we solve for the optimal hovering height of the UAV. Let χ^* be the optimal solution to $\mathbf{SP1-1}$ in (13). The problem for UAV hovering height optimization can be equivalently expressed as

$$\mathbf{SP1-2}: \underset{h_k}{\text{minimize}} \quad \lambda\left(\theta_{m_k}(h_k)\right)\left((\chi^*)^2 + (h_k)^2\right). \tag{14}$$
$$\text{s.t.} \quad H_l \leq h_k \leq H_u, \quad k \in \mathcal{K}. \tag{14a}$$

Through the above discussion, the 3-D optimal hovering positions of the UAV can be obtained by solving $\mathbf{SP1-1}$ and $\mathbf{SP1-2}$. ■

Proposition 2: $\mathbf{SP1-2}$ is a quasi-convex optimization problem.

Proof: Define $g(h_k) = \lg(\lambda(\theta_{m_k}(h_k))((\chi^*)^2 + (h_k)^2))$.

$$dg(h_k)/dh_k = z_1(h_k)/z_2(h_k) \tag{15}$$
$$z_1(h_k) = (\Delta_{\mathrm{LoS}} - \Delta_{\mathrm{NLoS}})\exp\left(-b\left[\theta_{m_k}(h_k) - a\right]\right) ab\sqrt{\chi^*}$$
$$+ 2h_k\left(1 + a\exp\left(-b\left[\theta_{m_k}(h_k) - a\right]\right)\right)^2/\ln 10, \tag{16}$$
$$z_2(h_k) = 10\left(1 + a\exp\left(-b\left[\theta_{m_k}(h_k) - a\right]\right)\right)^2\left(\chi^* + h_k^2\right). \tag{17}$$

TABLE 1. Values of the System Setting Parameters for Simulations

Parameter	Value	Parameter	Value
p	20 dBm	a, b	9.6, 0.28
σ^2	−174 dBm/Hz	η	1 Mb
Bandwidth	10 kHz	$c_{\min}, c_{\max}$	10, 40
$\Delta_{\mathrm{LoS}}, \Delta_{\mathrm{NLoS}}$	1.6 dB, 23 dB	H_l, H_u	50 m, 120 m

Let $z_1(h_k) = 0$

$$\frac{dz_1(h_k)}{dh_k} \geq \frac{2\left(1 + a\exp\left(ab - b\theta_{m_k}(h_k)\right)\right)^2}{\ln 10}\left(1 - \frac{b}{2}\right)$$
$$+ \frac{2h_k\left(1 + a\exp\left(ab - b\theta_{m_k}(h_k)\right)\right)}{\ln 10}\left(b\frac{d\theta_{m_k}(h_k)}{dh_k}\right) \geq 0 \tag{18}$$

$$\frac{d^2 g(h_k)}{dh_k^2} = \left(dz_1(h_k)z_2(h_k) - z_1(h_k)dz_2(h_k)\right)/z_2(h_k)^2. \tag{19}$$

We know from (17) and (18) that $z_2(h_k) > 0$, and $\frac{dz_1(h_k)}{dh_k} \geq 0$ when $z_1(h_k) = 0$. It means when $\frac{dg(h_k)}{dh_k} = 0$, $\frac{d^2 g(h_k)}{dh_k^2} \geq 0$. The function $g(h_k)$ is therefore quasi-convex. The optimum h_k^* can be obtain by using quasi-convex algorithms, such as the bisection method. ■

2) Adjusting the MTCD Clusters: Upon optimizing the UAV hovering position for every cluster, the optimal positions are designated as new cluster centres to adjust the clustering strategy. Specifically, for every MTCD, if switching from its current cluster to another cluster would shorten its data transmission time without violating the cluster size constraint or increasing the hovering time (being the farthest away MTCD) of the new cluster, this MTCD's cluster index should be changed.

B. INTERCLUSTER UAV FLYING PATH OPTIMIZATION

To achieve the overall optimization objective of problem **P1**, we next optimize the UAV's intercluster flying trajectory. Let $\mathbf{u}^*$ be the optimal hovering positions to **SP1** in (10). The corresponding subproblem is given as

$$\mathbf{SP2}: \underset{\boldsymbol{\beta}}{\text{minimize}} \quad \sum_{i=1}^{K}\sum_{j=1}^{K} \beta_{i,j}\varphi^f(\mathbf{u}^*)$$
$$\text{s.t.} \quad (9c), (9d), (9e). \tag{20}$$

Upon determining the optimal MTCD clustering and UAV hovering strategy as in Section IV-A, subproblem **SP2** is equivalent to a 3-D traveling distance minimization problem, i.e., the traveling salesman problem, which can be readily solved by the ACO algorithm.

V. SIMULATION RESULTS

In this section, we validate effectiveness and convergence of the proposed ACH–ACO algorithm in supporting delay-aware services. We first compare hovering time (data collection time) achieved by the ACH algorithm with the strategy based on k-means clustering. Then, performance of the ACH–ACO algorithm in reducing the overall mission time (hovering time plus traveling time) is compared with that achieved by random trajectories. Values of system setting parameters used in the simulations are summarized in Table 1.

A. Reducing UAV Hovering Time by ACH

In the simulations, the MTCDs are uniformly distributed in a square area with side length a. We first look at the convergence property

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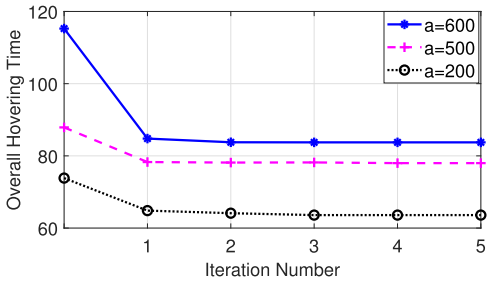

Fig. 2. Convergence of the ACH algorithm.

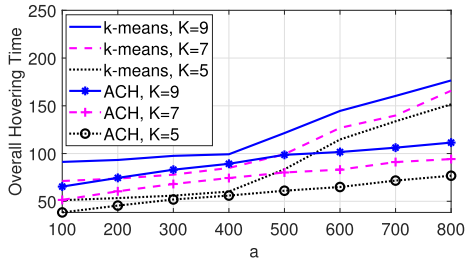

Fig. 3. Overall data collection time achieved by the ACH algorithm for joint MTCD clustering and UAV hovering position optimization.

of the ACH algorithm. The number of MTCDs and clusters are $M = 150$ and $K = 7$, respectively. The initial UAV flying height is set to a constant 80 m. Horizontal coordinates of the initial UAV hovering positions are set to that of the cluster centres selected by the k-means algorithm. By solving **SP1**, the optimal in-cluster UAV hovering positions, given the clustering strategy, are obtained. The clustering strategy is then adjusted using the optimal hovering positions as new cluster centres according to the greedy principle. Iterating the above alternating optimization process monotonically reduces the UAV's overall hovering time (data collection time) until convergence. Simulation results presented in Fig. 2 exhibit fast convergence of the ACH algorithm.

In less than five iterations, convergence of the ACH algorithm can be reached, and the corresponding MTCD clustering strategy and UAV hovering positions minimizing the overall hovering time can be obtained. The overall hovering time reduction increases with a, the side length of the MTCDs' distribution area. When $a = 600$ m, the ACH algorithm can reduce the UAV hovering time by about 27%.

Next, we compare the overall hovering time of the proposed ACH algorithm with the classic k-means algorithm. As in Fig. 2, we use $M = 150$ MTCDs and a fixed UAV flying height of 80 m, but with changing K and a values. The ACH algorithm was conducted for five iterations to obtain the results. It is observed from Fig. 3 that, for both strategies, the larger a is, the longer the overall UAV hovering time. Besides, the overall hovering time also increases with K, the number of clusters, as serving less MTCDs simultaneously result in lower spectral efficiency. Compared with the k means-based strategy, the ACH algorithm significantly reduces the time spent on completing the data collection tasks, especially when the MTCDs are distributed over a large area.

B. Overall Performance of the ACH–ACO Algorithm

The ACO algorithm is employed after the ACH process to find the shortest intercluster UAV flying route. Table 2 shows the performance of the ACH–ACO algorithm in terms of overall mission time. Compared with the strategy with k means-based clustering and ACO-based intercluster route planning, the ACH–ACO algorithm significantly reduced the overall mission time in all the cases in the numerical studies. In particular, the improvement is more significant when the task area and the number of MTCDs to be served are large.

TABLE 2. Overall Mission Time

Distribution Areas	MTCDs Number	Clusters Number	k-means-ACO	ACH-ACO	Delay Reduction
200	100	5	75.6 s	67.7 s	−10.4%
500	150	7	168.1 s	149.1 s	−11.3%
800	200	10	303.3 s	238.3 s	−21.4%

VI. CONCLUSION

Delay-aware design of UAV-enabled data collection for MTC networks has been studied, with a focus on clustering of the MTCDs and deployment of the UAV in the 3-D space. The total mission time minimization problem has been formulated as a challenging MINLP program. Based on the assumption of communication-only-when-hovering operation, the original problem has been decomposed into two tractable subproblems, which, respectively, deal with MTCD clustering and UAV hovering position optimization and intercluster flying path minimization. An ACH algorithm that alternately optimizes MTCD clustering and in-cluster UAV hovering positions has been proposed to solve the former subproblem, and the latter subproblem is readily solved by the ACO algorithm. Numerical results have shown quick convergence of the ACH process in obtaining the MTCD clustering and UAV hovering strategy that minimizes the hovering time. Total mission time of the proposed ACH–ACO approach has been examined and compared with the strategy based on k-means clustering in numerical examples. It has been observed that the ACH–ACO algorithm significantly outperforms the k means-based strategy in terms of total mission time, and the improvement is greater when the number of MTCDs and the mission area are larger.

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